UNIT - III

NON-UNIFORM FLOW IN OPEN CHANNELS

Subject: Hydraulics and Hydraulic Machinery (23A01404)

Syllabus at a glance

- Specific energy, critical flow, discharge curve
- Specific force, specific depth, critical depth
- Measurement of discharge and velocity
- Gradually varied flow; dynamic equation of GVF
- Hydraulic jump and classification; elements, characteristics; energy dissipation

1. Specific Energy

1.1 Definition

Specific energy E is the **energy per unit weight of water measured with respect to the channel bed (as datum)**.

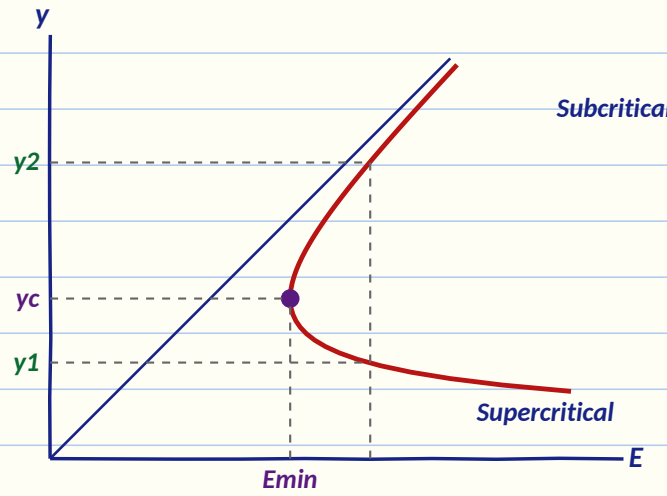
$$E = y + \alpha V^2 / 2g = y + Q^2 / (2 g A^2) \quad (\text{take } \alpha = 1)$$

- Total energy $H = z + y + V^2/2g$ changes along the channel because of the bed slope, but E changes only because of friction (E decreases in the flow direction when $S0 = 0$).

1.2 Rectangular channel (discharge per unit width q = Q/b)

$$E = y + q^2 / (2 g y^2)$$

- For a constant q, E is a **cubic** in y: for each $E > Emin$ there are **two positive roots (alternate depths)** y1 (supercritical, small depth, high velocity) and y2 (subcritical, large depth, low velocity), and one negative root.



$E = y + V^2 / 2g = y + q^2 / (2 g y^2)$

Two alternate depths (y1, y2) for one E; one at E = Emin

$y_c = (q^2 / g)^{1/3}$, $E_c = 1.5\, y_c$

1.3 Critical flow (minimum specific energy)

- Differentiate E with respect to y and put dE/dy = 0 (constant Q): $1 - Q^2 T / (g A^3) = 0$.

  $Q^2 T / (g A^3) = 1$ i.e. $Fr = 1$ or $A^3 / T = Q^2 / g$ (section factor $Z = Q/\sqrt{g}$)

- Rectangular channel results:

  $y_c = (q^2 / g)^{1/3}$; $V_c = \sqrt{(g\, y_c)}$; $E_c = 1.5\, y_c$; $(V_c^2/2g = y_c/2)$

- **Critical depth yc** depends only on discharge and channel shape, not on slope or roughness.

Channel	Critical depth relation	Ec
Rectangular	$y_c = (q^2 / g)^{1/3}$	1.5 yc
Triangular (side slope z)	$y_c = (2 Q^2 / (g z^2))^{1/5}$	1.25 yc
Trapezoidal / circular	By trial from $A^3/T = Q^2/g$	by calculation
Parabolic	-	1.333 yc

1.4 Classification by Froude number

State	Condition	Depth
Subcritical	Fr < 1	y > yc (V < Vc)
Critical	Fr = 1	y = yc
Supercritical	Fr > 1	y < yc (V > Vc)

1.5 Specific depth and critical slope

- **Specific depth** : the depth of flow corresponding to a given specific energy; the two **alternate depths** are the two specific depths for a given E.
- **Critical slope Sc** : bed slope at which uniform flow occurs at critical depth. For wide

rectangular channel (Manning): $S_c = g\,n^2 / y_c^{1/3}$.

- Mild slope ($S_0 < S_c$): $y_n > y_c$, subcritical uniform flow. Steep slope ($S_0 > S_c$): $y_n < y_c$. Critical slope: $y_n = y_c$.

2. Discharge Curve (Constant Specific Energy)

- For a fixed E: $q = y\sqrt{2g(E - y)}$. Plot q against y: q = 0 at y = 0 and at y = E, and passes through a **maximum**.

$$dq/dy = 0 \quad y = 2E/3 = y_c ; \quad q_{max} = \sqrt{g}\,(2E/3)^{3/2}$$

- So for a given specific energy the discharge is **maximum at critical depth**, and for a given discharge the specific energy is **minimum at critical depth**. A channel transition (hump, contraction) works on this principle.

Choking and effect of a hump or contraction

- For a rise in the bed (hump) of height Δz in subcritical flow, depth falls; E at the hump = $E1 - \Delta z$. If Δz becomes larger than E1 - Ec, flow is **choked**: upstream depth rises (backwater) so that critical flow occurs over the hump.
- Similarly, reducing the width b increases q; maximum q is reached when flow is critical at the contraction (**choking**).

Solved problems

Problem 1: Rectangular channel with $q = 2.5\ m^2/s$. Find yc, Vc and Ec.

$y_c = (2.5^2 / 9.81)^{1/3} = (0.637)^{1/3} = 0.86$ m; $V_c = 2.5 / 0.86 = 2.91$ m/s; $E_c = 1.5 \times 0.86 = 1.29$ m

Problem 2: Alternate depth: $q = 2\ m^2/s$, $y_1 = 0.5$ m. Find y2.

- $E1 = 0.5 + 4 / (19.62 \times 0.25) = 0.5 + 0.8155 = 1.3155$ m. Solve $y + 0.2039 / y^2 = 1.3155$ by trial: y = 1.15 gives 1.304; y = 1.17 gives 1.319.

$y_2 \approx 1.166$ m (subcritical, since $y_c = 0.742$ m)

Problem 3: Maximum discharge for E = 2 m (rectangular, per metre width).

$y_c = 2E/3 = 1.333$ m; $q_{max} = \sqrt{9.81} \times 1.333^{1.5} = 3.132 \times 1.540 = 4.82\ m^2/s$

Problem 4: Triangular channel z = 1.5, $Q = 1.5\ m^3/s$. Find yc.

$y_c = [2 \times 2.25 / (9.81 \times 2.25)]^{1/5} = (0.2039)^{0.2} = 0.727$ m (check: $Q^2 T / g A^3 = 1$)

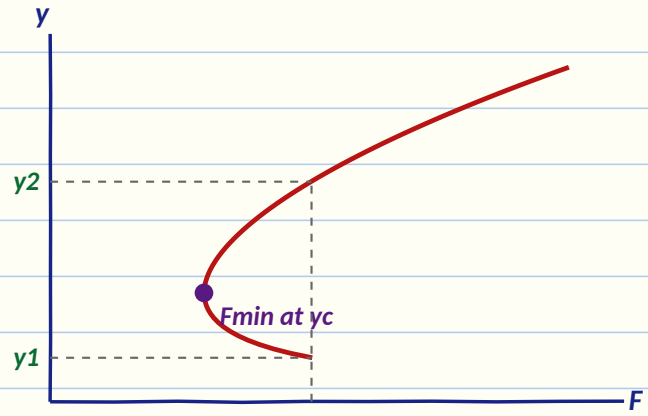
3. Specific Force (Momentum Function)

3.1 Definition

Specific force F = **(momentum flux + hydrostatic pressure force) per unit weight of water**, between two sections with no external force.

$$F = Q^2 / (g A) + A \bar{y} \quad (\bar{y} = \text{depth of centroid of area below water surface})$$

- Rectangular channel: $F = q^2 / (g y) + y^2 / 2$ (per unit width).



Specific force (momentum function):

$F = Q^2/(g A) + A \cdot \bar{y}$

Rectangular: $F = q^2/(g y) + y^2/2$

y1 and y2 = conjugate depths (equal F, hydraulic jump)

- The curve has a **minimum at critical depth**; for a given F there are two depths (**sequent or conjugate depths**) y1 and y2: same specific force but different specific energy. This is the principle of the hydraulic jump.

3.2 Specific energy vs specific force

Basis	Specific energy	Specific force
Based on	Energy equation (no loss assumed)	Momentum equation (no external force)
Two depths	Alternate depths	Conjugate (sequent) depths
Used for	Smooth transitions, gates, humps	Hydraulic jump, force on structures
Minimum	At critical depth	At critical depth

- In a hydraulic jump, F is conserved but E is lost.

Check using the jump data of Problem 8: q = 4, y1 = 0.5: F1 = 16 / (9.81 x 0.5) + 0.125 = 3.387; y2 = 2.316: F2 = 16 / (9.81 x 2.316) + 2.682 = 3.386 (equal).

4. Measurement of Discharge and Velocity

4.1 Velocity measurement

- **Current meter** (cup or propeller): n revolutions per second, v = a + b n (calibrated). **Pitot tube**, floats (surface, sub-surface, rod floats), ADCP, dye / salt dilution method.
- Mean velocity from one-point (0.6d) or two-point (0.2d, 0.8d) readings (Unit II).

4.2 Discharge measurement

Device	Discharge formula
Rectangular sharp-crested weir	$Q = (2/3)\, Cd\, \sqrt{(2g)}\, L\, H^{(3/2)}$
Francis formula (with end contractions)	$Q = 1.84\,(L - 0.1\, n\, H)\, H^{(3/2)}$
Triangular (V) notch	$Q = (8/15)\, Cd\, \sqrt{(2g)}\, \tan(\theta/2)\, H^{(5/2)}$
Broad-crested weir	$Q = 1.705\, Cd\, L\, H^{(3/2)}$ (max flow, critical depth on crest)
Venturi flume / Parshall flume	Throat at critical flow: $Q = Cd\, (2/3)^{(3/2)}\, \sqrt{g}\, b\, H^{(3/2)}$
Sluice (underflow) gate	$Q = Cd\, b\, a\, \sqrt{(2\, g\, y1)}$
Velocity-area method	$Q = \Sigma$ (width x depth x mean velocity)

- **Critical-depth meters** (weirs, flumes, Venturi flume) are reliable: Q depends on only upstream head.

Problem 5: Rectangular weir L = 2 m, H = 0.3 m, Cd = 0.62. Q = (2/3) x 0.62 x 4.429 x 2 x 0.3^1.5 = **0.60 m^3/s**.

Problem 6: 90° V-notch, H = 0.2 m, Cd = 0.6. Q = (8/15) x 0.6 x 4.429 x tan 45° x 0.2^2.5 = **0.0254 m^3/s**.

5. Gradually Varied Flow (GVF)

5.1 Meaning and assumptions

- GVF is **steady non-uniform flow in which the depth changes gradually along the channel length**, so that streamlines are nearly parallel and pressure is hydrostatic. Examples: backwater above a dam or weir, drawdown near a free overfall.
- Assumptions: **steady flow**; **prismatic channel**; hydrostatic pressure; small bed slope; friction slope Sf given by Manning / Chezy as in uniform flow with the local depth; α =

1.

5.2 Dynamic equation of GVF (derivation)

- Total head $H = z + y + V^2/2g$. Differentiate with respect to x: **$dH/dx = -Sf$** (energy slope), $dz/dx = -S0$.
- $dH/dx = -S0 + dy/dx + d(V^2/2g)/dx = -Sf$. Since $V = Q/A$, $d(V^2/2g)/dx = -(Q^2 T / g A^3)$ $dy/dx = -Fr^2\, dy/dx$.

$$dy/dx = (S0 - Sf) / (1 - Fr^2) \quad \text{(dynamic equation)}$$

- Using Manning for a wide rectangular channel with normal depth yn and critical depth yc:

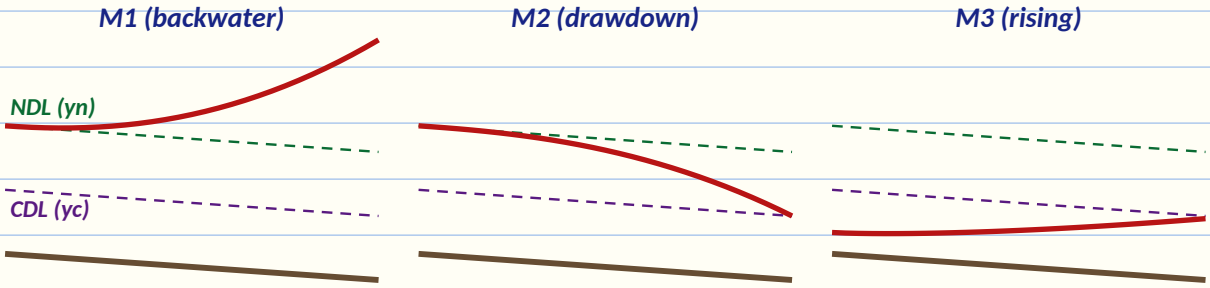
$$dy/dx = S0\,[\,1 - (yn / y)^{(10/3)}\,] / [\,1 - (yc / y)^3\,]$$

- **Numerator** : positive when $y > yn$ ($S0 > Sf$); **denominator** : positive when $y > yc$ (subcritical).

5.3 Classification of water surface profiles

- Zone 1: y above both lines; zone 2: between normal and critical; zone 3: below both. Slopes: Mild (M), Steep (S), Critical (C), Horizontal (H), Adverse (A).

Profile	Condition	dy/dx	Nature	Example
M1	$y > yn > yc$	+	Backwater (rising)	Dam on mild slope
M2	$yn > y > yc$	-	Drawdown (falling)	Approach to a free overfall
M3	$yn > yc > y$	+	Rising, ends in jump	Below sluice gate on mild slope
S1	$y > yc > yn$	+	Backwater	Mild-to-steep, behind a jump
S2	$yc > y > yn$	-	Drawdown	Break from mild to steep
S3	$yc > yn > y$	+	Rising	Below gate on steep slope
C1, C3	$y > yc$ (or $y < yc$)	+	Rising	Critical slope
H2, H3	$yc > y$ or $y > yc$	- or +	No normal depth ($S0 = 0$)	Horizontal bed
A2, A3	$y > yc$, $y < yc$	- / +	No normal depth ($S0 < 0$)	Adverse slope



Mild slope: yn > yc. M1: y > yn; M2: yc < y < yn; M3: y < yc

- Rule: **profiles in zone 2 are drawdown curves; in zones 1 and 3 they rise in the flow direction.** Near yn the profile approaches normal depth asymptotically; near yc it meets critical depth at a steep angle (dy/dx → infinity).

5.4 Computation methods

- **Direct step method** : $\Delta x = (E2 - E1) / (S0 - Sf)$; step depth-wise. Easiest for prismatic channels.
- **Standard step method** : for natural channels; trial and error on depth at the next section.
- **Graphical integration** , **numerical (Runge-Kutta)** , and **Bresse's method** (wide rectangular channels).

Direct step: $\Delta x = (E2 - E1) / (S0 - Sf)$, $Sf = (Sf1 + Sf2)/2$ with $Sf = n^2 V^2 / R^{(4/3)}$

5.5 Solved Problem 7 (backwater)

Rectangular channel b = 10 m, Q = 20 m^3/s (q = 2 m^2/s), n = 0.02, S0 = 0.001. A dam raises depth to 1.50 m. Find the distance upstream at which the depth is 1.40 m.

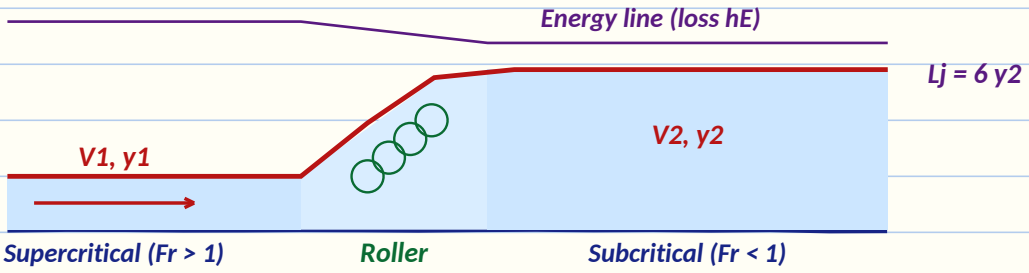
- yc = (4/9.81)^(1/3) = 0.742 m; wide channel yn ≈ (q n/√S0)^(3/5) = 1.15 m, so slope is mild and 1.5 > yn > yc: **M1 curve** .
- Section 1 (y = 1.5): V = 1.333; E1 = 1.5 + 0.0906 = 1.5906 m; R = 15/13 = 1.154 m; Sf1 = 0.0004 x 1.778 / 1.154^(4/3) = 5.88 x 10^{-4}.
- Section 2 (y = 1.4): V = 1.429; E2 = 1.4 + 0.1040 = 1.5040 m; R = 14/12.8 = 1.094 m; Sf2 = 7.24 x 10^{-4}.

Δx = (1.5040 - 1.5906) / (0.001 - 0.000656) = - 0.0866 / 0.000344 = - 252 m (about 252 m upstream)

6. Hydraulic Jump (Rapidly Varied Flow)

6.1 Definition

A hydraulic jump is the **sudden rise of water surface when a supercritical (rapid) stream changes into a subcritical (tranquil) stream** ; it is accompanied by **turbulence, a surface roller, air entrainment and large energy loss** .



- It occurs when the downstream depth (tailwater) is enough to sustain the **conjugate depth** y2 of the incoming supercritical flow, e.g. below a spillway, sluice gate or at a change from steep to mild slope.

6.2 Momentum equation (rectangular channel)

- Neglect bed friction and weight component; the **change in momentum flux = difference in hydrostatic forces** (per unit width):

$$\rho\, q\,(V2 - V1) = (1/2)\,\gamma\, y1^2 - (1/2)\,\gamma\, y2^2$$

$$y2 / y1 = (1/2)\,[\,\sqrt{(1 + 8\,Fr1^2)} - 1\,] \quad \text{and} \quad y1 / y2 = (1/2)\,[\,\sqrt{(1 + 8\,Fr2^2)} - 1\,]$$

- $Fr1 = V1 / \sqrt{(g\, y1)}$ must be > 1; (y1, y2) are **conjugate (sequent) depths**.

6.3 Elements (characteristics) of the jump

Element	Value / Formula
Height of jump	$hj = y2 - y1$
Energy loss	$hL = E1 - E2 = (y2 - y1)^3 / (4\, y1\, y2)$
Efficiency	$\eta = E2 / E1$ (ratio of downstream to upstream specific energy)
Length of jump	$Lj \approx 6\, y2$ (about 5 to 7 y2); roller length $Lr \approx 4$ to 5 y2
Power dissipated	$P = \gamma\, Q\, hL$
Relative energy loss	$hL / E1$ increases with Fr1 (about 45 to 70% at Fr1 = 4 to 9)

6.4 Classification of jumps (USBR, by upstream Froude number)

Fr1	Type	Description	Energy loss
1.0 - 1.7	Undular	Standing waves, no true jump	< 5%

1.7 - 2.5	Weak	Smooth rise, small rollers	5 - 15%
2.5 - 4.5	Oscillating	Unstable pulsating jet; waves travel downstream	15 - 45%
4.5 - 9.0	Steady (stable)	Well-balanced, stable, best for dissipation	45 - 70%
> 9.0	Strong (choppy)	Rough, high-velocity jet, intermittent slugs	70 - 85%

- Other classifications: by tailwater (**free, repelled, submerged jump**), jump on sloping floor, and **forced jump** (with baffles or sill).

6.5 Applications / uses

- **Dissipation of energy** below dams, spillways, weirs, sluices (prevents scour); **raising water level** downstream of a measuring flume; **mixing of chemicals** (water treatment); **aeration** of water; removal of air pockets in pipelines; maintaining high water level in irrigation channels.

6.6 Energy dissipation: stilling basins

- A **stilling basin** is a paved apron structure at the foot of a spillway in which the hydraulic jump is **contained** so that the high-energy flow is converted to low-energy flow before entering the river.
- USBR basins: **Type I** (plain, Fr1 < 2.5); **Type II** (chute blocks + dentated end sill, Fr1 > 4.5, V1 > 15 m/s); **Type III** (chute blocks + baffle blocks + end sill, V1 < 18 m/s, small structures); **Type IV** (for oscillating jump, Fr1 between 2.5 and 4.5).
- Accessories: **chute blocks** (break the jet), **baffle (friction) blocks** , **end sill** (reduces length of basin and guides flow), **tailwater depth** at least y2. **Roller bucket** and **ski-jump bucket** for high dams.

6.7 Solved Problem 8

A rectangular channel carries $q = 4\ m^2/s$ at $y1 = 0.5$ m. Find Fr1, y2, energy loss, efficiency, length and power dissipated per metre width.

- V1 = 4 / 0.5 = 8 m/s; **Fr1 = 8 / √(9.81 x 0.5) = 3.61** (oscillating jump).

y2 = (0.5 / 2) [√(1 + 8 x 13.05) - 1] = 0.25 x (10.265 - 1) = *2.316 m*

hL = (2.316 - 0.5)³ / (4 x 0.5 x 2.316) = 5.989 / 4.632 = *1.29 m*

- E1 = 0.5 + 64 / 19.62 = 3.762 m; V2 = 4 / 2.316 = 1.727; E2 = 2.316 + 0.152 = 2.468 m; check E1 - E2 = 1.294 m.
- Efficiency E2/E1 = 65.6%; Lj = 6 x 2.316 = **13.9 m** ; P = 9810 x 4 x 1.29 = **50.6 kW per metre width** .

Problem 9: Conjugate depth from y2

A jump forms with y2 = 3 m, V2 = 1.0 m/s. Fr2 = 1 / √(9.81 x 3) = 0.184; y1 = (3/2)[√(1 + 8 x 0.0339) - 1] = 1.5 x (1.1273 - 1) = **0.19 m** .

7. Applications and More Solved Problems

7.1 Control sections

- A **control section** has a definite depth-discharge relation (usually critical depth). Examples: **weir crest** , **free overfall** (end depth ≈ 0.715 yc), **break in grade from mild to steep** , **sluice gate** , **Venturi flume throat** .
- Subcritical flow is controlled from **downstream** ; supercritical flow is controlled from **upstream**. Computations of GVF start at a control section and proceed upstream (subcritical) or downstream (supercritical).

7.2 Flow over a hump (channel transition)

Problem 10: q = 2 m^2/s, upstream depth 1.2 m. Find the depth over a 0.15 m hump and the maximum hump height without choking.

- V1 = 1.667; E1 = 1.2 + 0.1416 = 1.3416 m; yc = 0.742 m; Ec = 1.113 m.

  Max hump (no choking) = E1 - Ec = 1.3416 - 1.113 = 0.229 m
- For Δz = 0.15 m: E2 = 1.3416 - 0.15 = 1.1916 m; y + 0.2039 / y^2 = 1.1916 gives **y ≈ 0.98 m** (subcritical, so surface falls over the hump).

7.3 Contraction in width

Problem 11: Q = 8 m^3/s, b1 = 4 m, y1 = 1.5 m. Find minimum width at which choking begins.

- V1 = 1.333; E1 = 1.5906 m. Maximum q at E1: q_max = √g (2E/3)^(3/2) = 3.132 x (1.0604)^1.5 = 3.42 m^2/s.

  b_min = Q / q_max = 8 / 3.42 = 2.34 m (narrower than this causes choking and backwater)

7.4 Sluice gate followed by a jump

Problem 12: Upstream depth 3 m, depth after the gate (vena contracta) y1 = 0.4 m. Find q, Fr1, y2 and energy loss (neglect approach velocity).

- 3 = 0.4 + $V1^2$ / 2g, so V1 = √(2 x 9.81 x 2.6) = 7.14 m/s; q = 0.4 x 7.14 = **2.86 m^2/s** .

- Fr1 = 7.14 / √(9.81 x 0.4) = 3.6; y2 = 0.2 (√(1 + 8 x 13) - 1) = **1.85 m** .

hL = (1.85 - 0.4)³ / (4 x 0.4 x 1.85) = 3.04 / 2.96 = 1.03 m

7.5 Direct step method: second step

- Continue Problem 7 ($q = 2\ m^2/s$, $b = 10\ m$, $n = 0.02$, $S0 = 0.001$):

y (m)	V	E (m)	R (m)	Sf x 10^{-4}	Mean Sf	Δx (m)	
1.50	1.333	1.5906	1.154	5.88	-	-	
1.40	1.429	1.5040	1.094	7.24	6.56	- 252	
1.30	1.538	1.4206	1.032	9.08	8.16	- 454	

- Total distance from the dam where y = 1.30 m = 252 + 454 = **706 m upstream** . Steps become longer as the curve approaches yn (1.15 m) asymptotically.

7.6 Jump relations for various Froude numbers (rectangular channel)

Fr1	y2 / y1	Energy loss hL / E1	Jump type
2	2.37	9%	Weak
3	3.77	26%	Oscillating
5	6.59	49%	Steady
9	12.24	70%	Strong

- **Rule** : a steady jump (Fr1 = 4.5 to 9) is the most efficient and most stable energy dissipator; avoid the oscillating range (Fr1 = 2.5 to 4.5) in design.

7.7 Difference between GVF and RVF

Basis	Gradually varied flow	Rapidly varied flow
Depth change	Slow, over long distance	Abrupt, over a short distance
Pressure	Hydrostatic	Non-hydrostatic, curved streamlines
Losses	Friction governs	Local turbulence governs (energy loss)
Examples	Backwater, drawdown	Hydraulic jump, flow at gate, over a weir

8. Quick Revision and Important Questions

$E = y + V^2/2g$. $yc = (q^2/g)^{(1/3)}$; $Ec = 1.5\ yc$; $Fr = 1$. q_{max} at $y = 2E/3$. $F = Q^2/gA + A\bar{y}$. $dy/dx = (S0 - Sf)/(1 - Fr^2)$. Jump: $y2/y1 = 0.5[\sqrt{(1 + 8Fr1^2)} - 1]$; $hL = (y2 - y1)^3/(4y1y2)$; $Lj = 6y2$.

- Define specific energy; draw the specific energy curve and derive the conditions for critical flow. (10 M)
- Define specific force and show that the specific force is minimum at critical depth. (5 M)
- Derive the dynamic equation of gradually varied flow. Explain the classification of GVF profiles. (10 M)
- Derive conjugate depth relations for a hydraulic jump in a rectangular channel. Obtain energy loss. (10 M)
- Classify hydraulic jumps. Describe the uses of the jump and stilling basins as energy dissipators. (5 M)
- Problems: critical depth, alternate depths, GVF by direct step method, jump height, loss and power.